

An Object-Oriented, Constraint-Based Heuristic for a Class of Passenger-Train Scheduling Problems

Mohammad T. Isaai and Madan G. Singh, *Fellow, IEEE*

Abstract—A constraint-based heuristic technique for predictive scheduling of passenger trains is presented in this paper. The algorithm, which is lookahead, has been designed using an object-oriented methodology and is suitable for a single-track railway with some double-track sections. It is shown that the heuristic tends to minimize total waiting time of the trains concerned. Simulation experiments with real data show the superiority of the schedules created to those by human experts with reference to data provided from predictive timetables and specifications of Iran Railways.

Index Terms—Constraint-based scheduling, constraint satisfaction problems, decision-support systems, heuristics, modeling, optimization, timetabling, train scheduling.

I. INTRODUCTION

TRAIN scheduling is an optimization problem that has been argued to be NP-hard [4], [9]. It was formulated for a single-track railway by Cai and Goh [4], using integer-programming techniques. The intractability of the problem was reported by analyzing an example. The problem consisted of 12 stations with the capacity for simultaneous accommodation of two trains, and nine trains traveling between end stations needed to be scheduled. The train's departure time from the origin was predetermined, and the speed of trains on each section of track was taken as constant. The decision variable, $X_{i,j}$, equaled one if train i stopped at station j and equaled zero otherwise. For such a problem, there could be up to $2^{108} \approx 3 \times 10^{32}$ possible solutions (many of which are infeasible) to be checked by exhaustive enumeration. Cai and Goh also showed that a special case of the train-scheduling problem is as hard as the knapsack problem, which is known to be NP-hard [4].

Train scheduling also has been considered as an optimization problem and has been tackled using analytical or OR-based techniques. Such models have provided valuable insights into the problem and are of theoretical interest. However, few models have solved "real-life" problems [10] and succeeded in practice. In this regard, the following factors are important.

- Analytical models demand enormous amounts of computation to find the mathematically guaranteed optimum solution.
- The modeling frameworks necessitate assumptions that simplify some aspects of the real problem to such an extent

that the theoretical aspects of the model dominate and weaken the practical one.

- It is not easy to exploit knowledge of the domain in the scheduling process.

The weaknesses of the analytical methods and the progress made in alternative approaches have led to the use of modeling techniques that generate good and acceptable solutions rather than optimal ones. Heuristics, simulation, and AI-based techniques are alternatives that have been applied to different classes of train-scheduling problems.

Constraint satisfaction is an AI-based approach that is close to OR techniques. This technique provides a much richer representational framework and facilitates the use of heuristic knowledge in solving the problem. Constraint-based techniques are incremental and extendable in contrast to "Black Box" techniques. They have been reported to have potential in solving various classes of scheduling problems. For the train scheduling problem, Chiu *et al.* [3] developed a constraint-based model to modify predictive train schedules disrupted by unforeseen events in practice.

This paper introduces a lookahead, constraint-based algorithm for predictive scheduling of passenger trains on a single-track railway with some double-track parts. Using the domain's knowledge, a heuristic has been developed and combined with a constraint-propagation technique similar to AC-5. Constraints are categorized into resources and temporal categories. Safety considerations concerning concurrent arrivals of opposite trains at the same station also are taken into account. The model has been developed using an object-oriented approach to facilitate expressing commonality among objects, modifying the algorithm, and extending the functionality of the basic computational representation of the problem if it is required [11]. Real predictive schedules, manually generated by train-planning experts for two corridors of Iran's railway, RAI, have been used to evaluate the model's outputs. Simulation experiments confirm that the algorithm is capable of creating excellent solutions rapidly for real-size problems in terms of total waiting time imposed on all trains involved.

II. PROBLEM REPRESENTATION

The problem consists of generating a predictive schedule for a set of passenger trains traveling on a (primarily) single-line railway that includes some double-line parts, using Iran's rail network and trains data. For any train being considered, the following data must be specified by experts: 1) its ready time for boarding at its origin station; 2) its minimum stop time for various reasons such as boarding and alighting at the origin and intermediate stations; and 3) its running time between all adja-

Manuscript received August 8, 1998; revised March 10, 1999. This work was supported in part by the Ministry of Culture and Higher Education, Iran.

The authors are with the Department of Computation, University of Manchester Institute of Science and Technology (UMIST), Manchester M60 1QD, U.K.

Publisher Item Identifier S 1094-6977(00)00363-1.

cent stations en route. The running time can be calculated using data from trains operations in the past.

The problem's decision variables are departure time at the origin, arrival time at the destination, and any of the above variables concerning any station en route for every involved train. Subsequently, stop time at each station is taken as a dependent variable with a value equal to or greater than the minimum planned stop time.

The constraints of the problem are as follows.

For each train t traveling between two adjacent stations (namely s and $s + 1$)

$$\text{Arrival}_{t,s+1} - \text{Departure}_{t,s} = \text{Running Time}_{s,s+1}.$$

For each train t at each station s en route

$$\text{Departure}_{t,s} - \text{Arrival}_{t,s} \geq \text{Planned Stop Time}_{t,s}.$$

Planned stop time is a constant time considered for boarding, alighting, or any other reason. If stopping is not required, it will be zero.

For any pair of trains $t1$ and $t2$ traveling in the same direction either on a single or a double-line track from station s to its immediate adjacent station $s + 1$

$$\begin{aligned} \text{Arrival}_{t1,s+1} - \text{Departure}_{t2,s} &\geq 0 \quad \text{or} \\ \text{Arrival}_{t2,s+1} - \text{Departure}_{t1,s} &\geq 0. \end{aligned}$$

For any pair of opposite trains $t1$ and $t2$ traveling on a single-line track between adjacent stations s and $s + 1$, if $t1$ is traveling from s to $s + 1$ (and $t2$ from $s + 1$ to s)

$$\begin{aligned} \text{Departure}_{t2,s+1} - \text{Arrival}_{t1,s+1} &\geq \text{Safety Interval} \quad \text{or} \\ \text{Departure}_{t1,s} - \text{Arrival}_{t2,s} &\geq \text{Safety Interval} \end{aligned}$$

where the safety interval is the time interval between consecutive occupation of a single-line track by opposite trains.

For each train t at its origin station o

$$\text{Departure}_{t,o} \geq \text{Ready Time}_{t,o} + \text{Stop time}_{t,o}.$$

To eliminate the possibility of collisions at stations, safety constraints are enforced. These constraints ensure that two opposite trains do not enter the same station at the same time (further discussion of this can be found in Section IV-A-1). Therefore, for any pair of opposite trains $t1$ and $t2$ approaching the same station s , either of the following constraints must be satisfied

$$\begin{aligned} \text{Arrival}_{t1,s} &\geq \text{Arrival}_{t2,s} + \text{Safety Interval} \\ \text{Arrival}_{t2,s} &\geq \text{Arrival}_{t1,s} + \text{Safety Interval}. \end{aligned}$$

Assuming that each station can accommodate trains using two kinds of tracks, namely "platforms" and "nonplatforms," for passenger trains stopping for boarding and alighting it is planned that they will use platforms for the comfort of passengers, and freight trains or trains stopping for other reasons are routed to other lines. However, generating a tight schedule for

a set of long-distance trains on a single-line railway may make violation of such a rule inevitable before even taking into consideration the consequences of unforeseen events in real life. In any case, for any station with p -platform and n -nonplatform lines at any point in time τ within the scheduling horizon

$$\begin{aligned} \text{The no. of trains stopped on platforms}_{s,\tau} &\leq p \\ \text{The no. of trains stopped on nonplatforms}_{s,\tau} &\leq n. \end{aligned}$$

In situations in which the above station's line-assignment rule has to be overridden, the number of stopped trains still must be less than or equal to $p + n$.

A. Solution Quality Criteria

The definition of a mathematical objective function can help measure the quality of any solution and thus help it to compare schedules generated by different algorithms. It also can be used for the comparison of computerized scheduling-system solutions with human experts' schedules. In addition, to solve an optimization problem, the comparison of the solutions would be a necessity for specifying direction of movements across the solution space. It is satisfied using an appropriate function. Considering the train scheduling as an optimization problem, we will compare the solutions obtained from the model with the experts' timetables. In this regard, three functions will be proposed in the following sections.

1) *The Ratio of Total Imposed Waiting Time to Train's Journey Time*: What is the meaning of "imposed waiting time" in this context? On a single-line track, if a section of track is engaged, a train requesting to enter the section has to stop at the prior station until the track becomes safely available. This time is called imposed waiting time. In a predictive train scheduling problem, where train-departure time and running times are predetermined, the aim of scheduling could be to distribute the imposed waiting time among trains involved with respect to an appropriate criterion.

One scenario is to schedule trains so as to minimize the average of unit waiting time (AUWT), where UWT is the ratio of the waiting time to minimum traveling time for each train. If so, the following objective function is suggested

$$\text{Minimize} \left\{ \text{Average}_{\text{Trains}} \left(\frac{\sum_{\text{Stations}} \text{Waiting Time}}{\sum_{\text{Lines}} \text{Running Time} + \sum_{\text{Stations}} \text{Planned Stop}} \right) \right\}.$$

Function 1

The drawback of an average is the possibility of generating very good schedules for most trains at the expense of generating very bad schedules for a few others. If, for example, allocating the same amount of waiting time to a long-distance and a short-distance train is undesirable, and the problem includes

both categories of trains, the following objective function outperforms Function 1.

$$\text{Minimize} \left[\text{Max}_{\text{Trains}} \left(\frac{\sum_{\text{Stations}} \text{Waiting Time}}{\sum_{\text{Lines}} \text{Running Time} + \sum_{\text{Stations}} \text{Planned Stop}} \right) \right].$$

Function 2

For any given schedule, the ratio is calculated for each involved train. Then the maximum figure is treated as the schedule's quality (to be referred to as MRWJ). The lower the maximum ratio, the higher the quality of the schedule.

2) *Total Weighted Imposed Waiting Time (TWT)*: In the real world, trains are of different value in the eyes of passengers and managers of a railway system, as facilities provided to passengers inside trains, speed, and fares vary. Therefore, each train can be assigned a weight representing its importance within the railway system. The objective of a scheduling system may be to reduce the total weighted imposed waiting time (TWT) for all trains involved

$$\text{Minimize} \sum_{i=1}^{\text{Trains}} \left(\text{Weight}_i * \sum_{\text{Stations}} \text{Imposed Waiting time} \right).$$

Function 3

A similar function was used in the [2] model, where each stop, movement, departure, and arrival time of a train was assigned an individual cost. Although Carey's objective function was theoretically more precise than Function 3, it is not clear to the authors that, in real environments, such a system would work due to the excessive number of weights (costs) needed to be assigned.

III. MODEL COMPONENTS

Three modules of the model that collaborate to generate the solution will be explained in this section. The first module, which is called "initialization," deals with the creation of computational structures and data inputs for the problem at hand. The second module generates a feasible solution using a heuristic. This module is called "scheduling." Finally, the evaluation module computes the quality of the solution using one of the three functions proposed in Section II-A.

A. Initialization

The basic structure on which the heuristic will be developed is an object-oriented interpretation of trains and the rail network. The two categories of objects in the train-scheduling problem are resources and trains.

Resources consist of those stations that are not placed at junctions and those that link branches of the rail network to each

other. The terms "ordinary stations" and "junctions" are used to refer to each of these respectively. Each part of the track connecting two adjacent stations is called a "block." Blocks are either single-track or double-track, and the network mainly consists of single-track blocks.

Trains use available capacity of resources to carry passengers. Each part of the journey, which is either a movement on a block or stop/waiting at a station, is referred to as an "activity." Activities relating to a train are linked together so that the sequence of activities resembles that of the resources on which the train operates.

The initialization phase is divided into two stages. The first stage deals with creating and linking objects of blocks and stations based on Iran's rail network design. The second stage creates objects of trains, a link-list of activities for each train based on train-related data and the network created in the first stage.

For the scheduling problem, the decision variables are the attributes of the activities representing operations through which trains connect their origin and destination. This will be further discussed in the following section.

B. Scheduling

The scheduling process assigns departure and arrival times to the involved trains at their origins, destinations, and intermediate stations. To meet the goal, a constraint-based heuristic has been developed and tested. In this regard, Section IV categorizes the constraints concerned. Section V concentrates on variable and value ordering and on variable and value-selection criteria. The constraint-propagation technique used will be discussed in Section VI.

"Start" and "finish" times are two variables associated with each activity, which represents either movement of a train on a block or a stop at a station. The decision variables of the scheduling problem are start and finish of those activities that represent train movements on blocks. In general, the start time of an activity that is executable on a station is equal to the finish time of its preceding activity, and its finish time is equal to the start time of its succeeding activity, in which both preceding and succeeding activities are executed on immediately adjacent blocks in the neighborhood of the station. Therefore, the number of variables associated with a train passing through m stations including its origin, destination, and intermediate stations is twice as many as the number of blocks involved, i.e., $2 \times (m - 1)$.

C. Evaluation

Once the scheduling process ends, the quality of the schedule obtained is calculated using the selected criterion. There are three options available to the user (the three functions introduced in Section II-A). The evaluation stage is used to compare the model's output with the benchmark concerned.

IV. CONSTRAINTS

A feasible solution is a schedule of train operations that satisfies all constraints. Constraints can be used to either test the validity of the solution or reduce the computational effort needed to solve the problem [1]. Considering the scheduling as a search process, the latter approach aims to reduce the search space and

consequently the time required for finding an appropriate solution. In this model, constraints are used not only to test the validity of the decisions made in the scheduling process but also to prune the domain of the variables. Section VI, which follows, concentrates on the constraint-propagation technique applied to the problem at hand.

Constraints concerning the train-scheduling problem can be categorized into temporal and resource constraints. The former represent precedence relationships between activities, while the latter deals with the availability of resources (i.e., blocks and stations). Safety, the paramount factor in railways, is also incorporated in the modeling of the problem in the two methods: a category of temporal constraints and a constant parameter in resource constraints associated with single-track lines.

The following terms are defined to facilitate discussion on constraints affecting the execution of activities:

- r a resource (block or station);
- t_r number of trains demanding r ;
- $A_{t,r}$ an activity relating to train t executable on r ;
- $D_{t,r}$ traveling direction for train t on resource r (1 or 0);
- A_r set of activities demanding r ;
- $A_r = \{\forall A_{t,r}, t = 1, \dots, t_r\}$;
- $U_r = \{\forall A_{t,r}, t = 1, \dots, t_r, D_{t,r} = 1\}$;
- $D_r = \{\forall A_{t,r}, t = 1, \dots, t_r, D_{t,r} = 0\}$;
- $A_r = U_r \cup D_r$.

A. Temporal Constraints

Different types of precedence relationships linking any two activities together are called temporal constraints. Instances of such constraints are the equality/inequality relationship between two variables representing the start and finish times of two activities with respect to the minimum time interval between the two. When only temporal constraints are considered, constraint propagation is sufficient to compute earliest start and finish times of activities and to determine whether a set of temporal constraints is consistent [8]. Nevertheless, resource constraints are involved in the scheduling problem as well.

For the problem at hand, two categories of temporal constraints are those linking activities of the same train and those representing connections between passenger trains. For instance, Fig. 1 shows a train traveling between two neighboring stations. The train's operations are broken into three activities, and their precedence relationships, together with a temporal constraint graph, are shown in Fig. 2.

It is worth mentioning that duration (A), duration (C), and ready time (A) are constants showing minimum stop times at stations M and N and running time on the block (M to N), respectively. The "Scheduling Horizon" is also a constant.

1) *Safety Constraints*: The regulations of rail transport in Iran's railways bans concurrent arrivals of two opposite passenger trains at the same station [5, article 38] with one exception [5, article 38, pt. 11.2]. This aims to eliminate the chance of multiple allocation of station lines due to human errors and to avoid simultaneous monitoring of arrivals to prevent collisions.

Therefore, a minimum time interval between the arrivals is required, and this is referred to as the "Safety Interval" in this paper. Although the safety interval is actually a real figure determined by the train's speed and length, in practice, train dis-

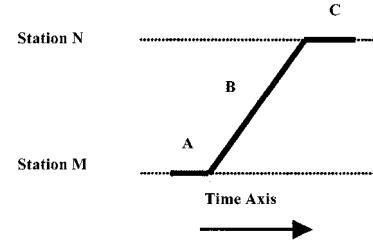


Fig. 1. Simple distance-time graph.

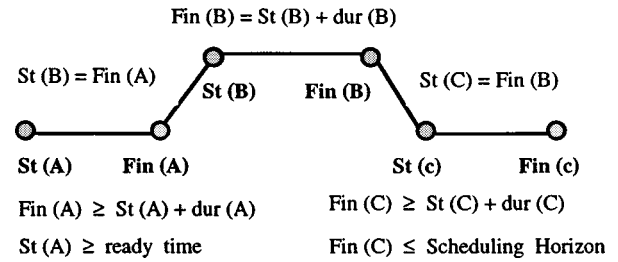


Fig. 2. Temporal constraints of the example.

patchers work with integer figures, which are multiples of a minute. Using the data available to the authors, the safety interval is fixed at one minute. Assuming activities $A_{t1,r1}$ and $A_{t2,r2}$, $r1 \neq r2$ represent movements of opposite trains traveling on different lines next to a station, either of the following constraints (1) must be satisfied.

$$\begin{aligned} \text{Finish}(A_{t1,r1}) &\geq \text{Finish}(A_{t2,r2}) + \text{Safety Interval} \\ \text{Finish}(A_{t2,r2}) &\geq \text{Finish}(A_{t1,r1}) + \text{Safety Interval}. \end{aligned} \quad (1)$$

For an activity A there is a set of activities $B = \{B_j, j = 1, \dots, n\}$, for which the safety constraints must be satisfied. Propagation of such constraints means $n \times (n + 1)/2$ comparisons need to be made of A 's earliest finish with B_j 's. For the case where the earliest finish times are equal, propagation will take place based on the second part of (1) (i.e., the domain of B_j 's earliest finish time will be pruned).

B. Resource Constraints

Resource constraints are categorized into station-related and track-related. The latter incorporates single-track and double-track constraints. They are used to avoid infeasible allocation of activities to resources and also to reduce domains of variables where necessary. The following sections detail the use of disjunctive constraints in connection with tracks and design of timetables for handling capacity of stations.

1) *Track-Related Constraints*: In Iran's railways, a block (the track between adjacent stations) is either single or double-line. A single-line block provides capacity to trains traveling in both directions, whereas, each line of a double-line track is associated with one direction only. Consequently, a single-line block and each line of a double-line block are considered as unary resources (resources with a capacity of one train at a time).

There are two methods concerning assignment of a unary resource [7]. The first method, cumulative representation, states that the capacity required by all activities at any point in time

cannot exceed the capacity available at the time. Since capacity varies with time, at any time, the resource may or may not be available. In contrast, the second method, disjunctive representation, states that for any pair of activities A and B (which require the same unary resource), either A must precede B , or B must precede A .

If the disjunctive representation is used, many pieces of information can be deduced. For instance, it is sufficient to decide/prove that A should not/cannot precede B , to deduce that B must precede A . For the problem at hand, at each stage of the scheduling process, an activity is selected using a heuristic and assigned to the relevant resource. Consequently, any other overlapping activity requiring the same resource will succeed the activity.

a) *Single-Track Constraints:* Safety is a crucial subject concerning rail services, especially safety of passenger services. To ensure safety in the real world, only one train is allowed to occupy a single-track block at any time, where the block is not divided into sections via a signaling system. Moreover, a time interval between occupation of the line by two consecutive activities representing opposite trains is considered. Thus, single-track constraints are dealt with in two parts. First, for any pair of activities ($A_{t1,r} \in U_r$, $A_{t2,r} \in D_r$) relating to opposite trains, a safety interval is considered to avoid either head-on collisions on the block or head-to-tail collisions at the station's entrance area. Therefore, the following disjunctive constraints are added to the model:

$$\begin{aligned} \text{Start}(A_{t1,r}) &\geq \text{Finish}(A_{t2,r}) + \text{Safety Interval} \quad \text{or} \\ \text{Start}(A_{t2,r}) &\geq \text{Finish}(A_{t1,r}) + \text{Safety Interval}. \end{aligned} \quad (2)$$

In what follows, the safety interval is a constant integer variable equal to one minute. It means that a train is permitted to depart the station safely once all coaches forming the approaching opposite train have entered the station's area.

For any pair of activities ($A_{t1,r} \in U_r$, $A_{t2,r} \in U_r$) or ($A_{t1,r} \in D_r$, $A_{t2,r} \in D_r$) the safety interval is not needed. In other words, once the train, which has departed from station $S1$, arrives at station $S2$, the next train is allowed to depart at $S1$ toward $S2$. Then, either of following constraints must be satisfied:

$$\begin{aligned} \text{Start}(A_{t1,r}) &\geq \text{Finish}(A_{t2,r}) \\ \text{Start}(A_{t2,r}) &\geq \text{Finish}(A_{t1,r}). \end{aligned} \quad (3)$$

For a selected activity $A_{t1,r}$ for scheduling on a single-track block r , propagation of track-related constraints starts with $t_r \times (t_r - 1)/2$ checks, where t_r is the number of activities demanding the block. In case any activity overlaps $A_{t1,r}$, the domain of its earliest start will be pruned based on the second part of constraints (2) and (3).

b) *Double-Track Constraints:* For a double-track block, each line is associated with only one direction, so that opposite trains, requiring the same block, are planned to run on different lines. Indeed, activities relating to opposite trains do not compete with each other since they are assigned to different unary resources. Accordingly, constraints (3) are applied to activities demanding the same line of a double-track as well.

For a selected activity $A_{t,r} \in D_r$ (or U_r) to schedule on a double-track block r , the propagation of track constraints starts with $S_r \times (S_r - 1)/2$ checks, where $S_r \leq t_r$ is the number of activities in D_r (or U_r). In case any activity overlaps the selected one, the domain of its earliest start will be pruned based on the second part of constraint (3).

2) *Station-Related Constraints:* In Iran's railway, stations have different kinds of lines to accommodate a variety of trains with different aims of arrival at stations. Reference [5] describes the lines and the kinds of services for which they are considered. Two categories of station lines involved in predictive scheduling of passenger trains are those allocated to accommodating arriving and departing trains. Passenger trains stopped for boarding and alighting need platform lines to facilitate passengers' movement. On the other hand, assuming unavailability of an immediate intermediate track causes a delay to a train, the train does not have to occupy a platform line. Such a train is directed to an alternative line.

For a given station, typical constrained-capacity variables are P_τ and NP_τ , denoting the number of platform and nonplatform lines, respectively, available at any point in time τ . In the model for any station, a timetable is generated to update available capacity and prevent violation of the maximum capacity. Each element of the timetable is an object with three members: the values assigned to P_τ , the values assigned to NP_τ , and event kind (arrival or departure). There are two different procedures concerning timetable generation. The first approach keeps and updates values of the variables at any time unit (e.g., every minute, without regard to occurrence of events). The alternative timetable does the same but for the points in time at which an event occurs. Assuming that one object is created for each element on the timetable, the former approach requires more space for keeping the information and more time when iterating the table is needed. Therefore, the second approach was selected in this research.

Scheduling an activity on a block leads to updating the timetables of stations existing at both ends of the block. On the other hand, unavailability of station capacity leads to updating the domains of activities. In other words, constraint propagation is bidirectional (from resources to activities and vice versa).

V. VARIABLE ORDERING AND SELECTING

The variables of the problem are the activities' start and finish times. Each variable is an attribute of the activity concerned. Therefore, ordering and selecting activities convey the same meaning as those of variables. Any train is broken into a set of activities that are linked together (Fig. 4). Activities are linked to resources. For example, the first three activities of a given train are linked to the origin station, the first block, and the first intermediate station, respectively. Activities of the same train are also linked together resembling a chain. The position of an activity on the chain is the same as that of the resource concerned with the train's route. In what follows, ordering, selecting, and rejecting of variables only refer to those variables that are attributed to activities linked to blocks and not to stations. A block-related activity is linked to two station-related activities. Finish time of the prior activity is the same as start

and start time of the following activity is the same as finish time of the activity.

A. Nominated-Variables Set

At each stage of the scheduling process, a set of nominated variables is used to pick a variable for value assignment. The set includes n variables at most, where n is equal to the number of trains involved. Each train has merely one representative in the set as long as there is any unscheduled activity among its activities. The representative of a train is the variable whose immediate predecessor has been selected and assigned a value at previous stages. As the scheduling process proceeds, the number of variables in the set is reduced along with a reduction in the number of unscheduled trains. Finally, the scheduling process will end if the nominated variable set is empty ($n = 0$).

B. Variable-Selection Criteria

The aim is to select a variable from the set of nominated variables using the problem knowledge in the hope of generating an appropriate solution. Accordingly, the variable with minimum earliest finish (MEF) time is selected. In case the MEF is not unique, the subset of nominated variables set concerned is called S_{MEF} . If activities in S_{MEF} do not compete for the same resource, the variable in S_{MEF} , which was found first in the process, will be selected. Otherwise, if any variable in S_{MEF} competes with the one that was found first, the variable with minimum earliest start (MES) will be selected. In case MES is not unique, the variable that was found first in the process will be selected.

The heuristic is not a direct conflict-resolution technique, because it does not concentrate on locating and resolving resource contentions. However, in this section, a conflict-oriented interpretation will be used to show the capability of the heuristic in implicitly resolving conflicts on blocks. In fact, selecting variables based on the above criteria minimizes local delays in four out of five types of resource contentions that are likely to occur on blocks.

For a typical resource contention including activities $A1$ and $A2$ with running times $R1$ and $R2$, five different scenarios and decisions made by the algorithm are as follows.

- 1) Schematic diagram (a) illustrates two competing activities for which $ES(2) > ES(1)$, $R1 = R2$ and $EF(2) > EF(1)$. Local waiting time (LWT) corresponding to the alternative decisions is

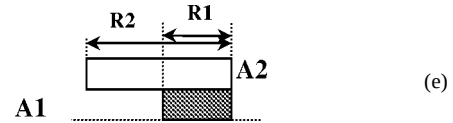
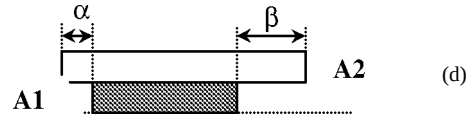
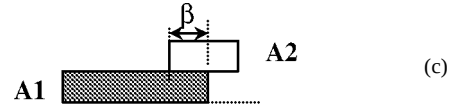
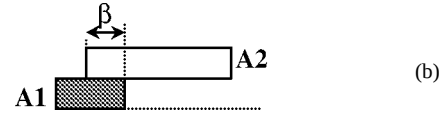
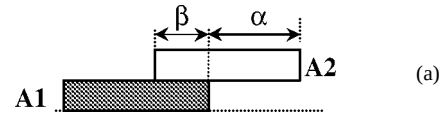
$$LWT = \begin{cases} \beta < R2, & \text{if } A1 \text{ is scheduled first;} \\ R1 + \alpha > R2, & \text{otherwise.} \end{cases}$$

The algorithm selects $A1$ and hence, LWT is minimized.

- 2) Two competing activities for which $ES(2) > ES(1)$, $R1 < R2$ and $EF(2) > EF(1)$ are illustrated in diagram (b). For such a case

$$LWT = \begin{cases} \beta < R1, & \text{if } A1 \text{ is scheduled first;} \\ R1 + R2 - \beta > R1, & \text{otherwise.} \end{cases}$$

Giving priority to $A1$, the heuristic minimizes LWT.



- 3) Diagram (c) shows a resource contention for which $ES(2) > ES(1)$, $R1 > R2$ and $EF(2) > EF(1)$

$$LWT = \begin{cases} \beta < R2 < R1, & \text{if } A1 \text{ is scheduled first;} \\ R1 + R2 - \beta > R1, & \text{otherwise.} \end{cases}$$

Selecting $A1$, the heuristic minimizes LWT.

- 4) For the given pair of activities, if $ES(1) > ES(2)$, $R1 < R2$, $EF(1) < EF(2)$, then

$$LWT = \begin{cases} R1 + \alpha, & \text{if } A1 \text{ is scheduled first;} \\ R1 + \beta, & \text{otherwise.} \end{cases}$$

The algorithm selects $A1$ with respect to the variable-selection rule and without regard to α and β . If $\alpha < \beta$, LWT is minimized (diagram d).

- 5) Assuming $ES(1) > ES(2)$, $R1 < R2$ and $EF(1) = EF(2)$, $A2$ is selected before $A1$. Consequently, LWT is minimized (diagram e)

$$LWT = \begin{cases} R2, & \text{if } A1 \text{ is scheduled first;} \\ R1 < R2, & \text{otherwise.} \end{cases}$$

It is worth noting that delays imposed for safety assurance are excluded from the LWT, since such delays increase LWT without influencing the minimizing choice. In fact, such a delay is not required when competing activities represent trains traveling in the same direction. In case of resource contention by opposite trains, delay for safety increases LWT by the amount equal to the safety interval without regard to which train runs first.

C. Confirming Variable

How does the algorithm finalize variable selection? We recall that the selected activity is executable on a block, and the next activity with respect to the same train will be executed on a station. The algorithm checks availability of an appropriate line in the concerned station at the time execution of the picked activity is finished. If a suitable line is ready, the activity will be scheduled. Otherwise, it will be rejected. The consequences of the rejection will be described in Section V-D. The rationale regarding the mentioned procedure is the train-dispatching mechanism in the real world, which is as follows.

In real environments within a single-track railway, a train departs a station only if the next intermediate station is able to accept the train safely and reserve a suitable line for that [5, articles 36, 57.2], [6, articles 100.4, 123.2]. If the train is scheduled to stop at the station for boarding and alighting, an idle platform is required. For stopping for other reasons, a nonplatform line must be available. Likewise, transient trains must traverse an idle station line before entering the next block. Consequently, in case a station is not able to accept a coming train at a point in time, the train's departure time at the previous station will be postponed with reference to the new arrival time at the station.

D. Rejecting Variable

In case of variable rejection, the domain of the rejected variable will be reduced. In reality, this is equivalent to postponing the departure of the concerned train at its current station to such an extent that it can be directed to an appropriate line of the next station at its arrival time.

Having reduced the domain of the rejected variable, the algorithm focuses on the following two steps:

- triggering constraints propagation processes to reduce the domain of the relevant variables;
- repeating the variable-selection process.

Reducing the domain of any rejected variable is performed with reference to the following:

- the type of station line suitable for a stop;
- the type of track (double or single) to which the rejected activity is linked;
- duration of safety interval;
- traveling direction attributed to the rejected activity;
- the number of available alternative lines.

Rejecting any variable is considered as a case for which a procedure is followed to determine the new earliest start of the rejected variable. In this regard, a set of if-then rules have been defined to specify the new earliest start with reference to the state of the mentioned facts. This part of the model is referred to as "Rule-Based Case Handling" in Fig. 3.

VI. CONSTRAINT PROPAGATION

In the model, AC-5 is the constraint-propagation technique that has been used to update the lower boundaries of the variables. Start and finish are two variables associated with each activity, for which the domain boundaries are the earliest start, latest start, earliest finish, and latest finish. Having specified the scheduling horizon for a given problem, latest start and latest

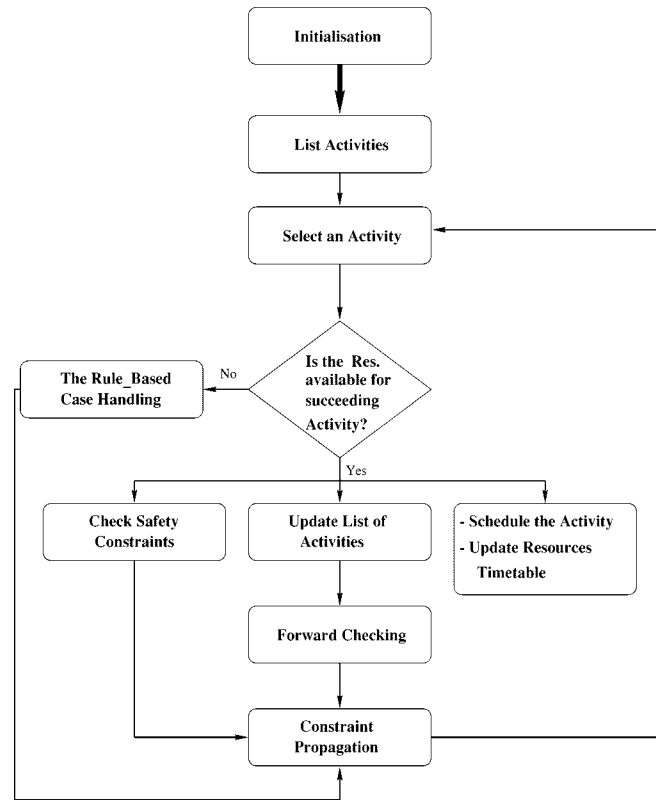


Fig. 3. Structure of the lookahead algorithm.

finish are calculated before scheduling starts. Then, at the scheduling stage, the propagation process only updates the earliest start and earliest finish of activities. It is a bidirectional process, from temporal to resource constraints and vice versa.

The algorithm is lookahead, and backtracking is not required. Following the scheduling of an activity *A* on its relevant block, all activities competing for the same resource are checked for resolving resource contentions possibly occurring over the time interval allocated to *A*. The domain of such activities is reduced to such an extent that they will be scheduled after *A*.

VII. COMPUTATIONAL REPRESENTATION

The implementation aspects of a decision-support system are very important, as they may affect efficiency of the algorithms used. For the problem at hand, an object-oriented methodology along with Borland C++ version 4.0 language was used to facilitate design, programming, and modification of the computerized system.

Trains, activities, stations, and blocks are objects of different classes, created at the running stage using a "new" operator for providing a flexible platform that is extendible to wider applications such as exploring the effect of adding or removing stations on a given timetable. Ordinary (station) and block are derived classes of a public base class called resource. "Junction" is also a derived class of ordinary. In fact, inheritance is used to facilitate expressing commonality among classes [11] and linking objects of different classes for simulating Iran's rail network.

Fig. 4 shows that, for any object in the list of trains, there is a double-linked list of activities, one for each resource en route between the origin and destination. Each activity keeps

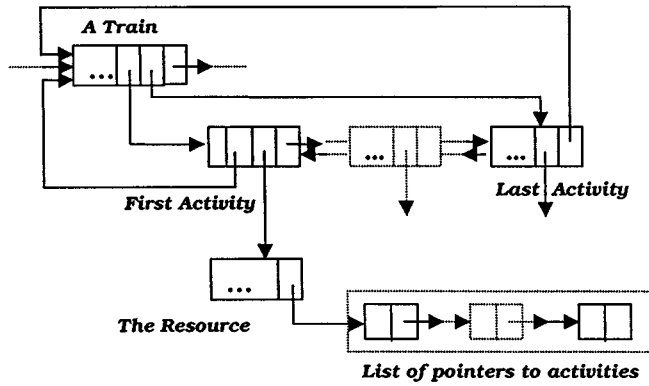


Fig. 4. Links between different objects.

the address of a resource, and similarly, each resource is linked to a list of objects holding addresses of activities executable on the resource.

The underlying part of the initialization phase is the automatic generation of activities for any train involved. To do so the network first is divided into a set of branches, in which a branch covers stations and blocks between either two neighboring junctions or a junction and the end station of a route. Each block and station is associated merely with a branch, and the branch is assigned a unique code. Then, an object-oriented list of routes is generated to specify, for any two given branches, the sequence of branches en route as well as traveling direction on the concerned branch. This information is used for creating activities, associating traveling direction to them, and establishing the link between activities and resources for all trains concerned.

The constraints of the problem are of the binary kind, including only two variables for which a constraint is either an equality or an inequality relationship between the variables. A typical variable can be either a start time or a finish time of an activity. Hence, a constraint can formulate dependency of either two activities or attributes of the same activity to one another. The object-oriented approach to constraints leads to the removal of constraints storage, since a constraint is an object of a certain class. It is generated when it is required and removed when it is not useful to save computer memory.

In connection with constraint representation, inheritance is used once more to make the most use of commonality between constraints in favor of the algorithm's efficiency. For example, "temporal" is the abstract class from which different classes are derived, one class for each sort of constraints. Each derived class contains only a member function that reduces domains of variables involved in a particular binary constraint, according to AC-5 technique. These functions are, in fact, a redefinition of a pure virtual function from the private base class temporal. The object-oriented approach also is used to implement the AC-5 algorithm that was originally developed by [12].

VIII. EXPERIMENT RESULTS

The passenger-train scheduling model has been tested using the data of two corridors of Iran's railways. In fact, the heuristic and other components of the scheduling process were run in connection with a computational foundation representing a

set of stations and tracks linked together based on Iran's rail network. Data are timetables of passenger trains, generated by train-planning experts for two corridors: Tehran–Mashad and Tehran–Tabriz, connecting the capital to two major cities in the Northeast and Northwest of the country, respectively. The two routes are transit lines connecting Turkey in the Northwest to Turkmenistan in the Northeast.

Cases P1, P3, and P5, introduced in Table I, relate to the Tehran–Mashad route, on which ten out of 50 blocks were double-track at the time of timetable (data) publication, and the remainder were single-track. The second line is under construction and extension rapidly, to fulfill the increase of passenger and transit demand on the route. This fact has been a motivating factor regarding developing the model for single and double-track rail networks. In addition, cases P2 and P4 relate to the Tehran–Tabriz route, which is entirely single track. Table I provides an insight into the specifications of the rail corridors and trains involved.

Basically, the model is an advisory system for train scheduling experts. For any train, departure time at origin station and minimum time for boarding and alighting at stations en route are both specified by experts. We note that such subjects relate to passenger-demand pattern and the policy of the corporation that are out of the scope of this research. To solve the cases at hand, running time between neighboring stations is considered as constant, taken from the real, planned timetables. In practice, average running time for each category of trains at both directions can be used.

Three functions described in Section II-A are used to evaluate the quality of schedules generated by the model. In addition, the quality of schedules generated by the expert train planners and the heuristic are compared in Tables II–IV.

Following the discussion in Section V-B on the tendency of the algorithm to minimize total LWT, the model was tested with reference to the TWT criterion, (Function 3), in which all trains have been assigned equal weights ($=1$). From a mathematical viewpoint, the results show that the total waiting time has decreased by between 17.68% and 27.34% (Table II). The quality of the generated schedule for all problems is superior to those for expert planners. However, the least improvement relates to P1. Thus, the heuristic competes with expert schedulers in minimizing the total LWT of all trains concerned. In addition, the quality of the algorithm's schedules in terms of AUWT shows that the model has yielded improvements between 11.21% and 27.41% (figures are shown in Table III). Table IV shows that improvements achieved in terms of MRWJ are between 1.90% and 21.91%

IX. CONCLUSION

The output of the scheduling algorithm in terms of TWT imposed on trains is satisfactory. In addition, the results of the model with respect to other objective functions are promising as well.

It should be noted that it would be advantageous for the advisory system to be free from limitations such as objective functions. This is a goal of future research.

TABLE I
SPECIFICATIONS OF THE CASES SOLVED BY THE HEURISTIC

Cases	Trains			Activities		Blocks			Stat-ions	Route (km)
	Up*	down*	Total	track	Total	single	Double	Total		
P1	11	11	22	932	1886	40	10	50	51	926
P2	8	8	16	596	1208	47	0	47	48	736
P3	8	9	17	850	1717	40	10	50	51	926
P4	5	5	10	456	922	47	0	47	48	736
P5	8	8	16	637	1290	36	5	41	42	795

TABLE II
COMPARISON ON TWT MEASURE

Case:	P1	P2	P3	P4	P5
Benchmark	441	230	362	139	246
Heuristic	418	200	298	101	217
Improved %	5.22	13.04	17.68	27.34	11.79

TABLE III
COMPARISON ON AUWT MEASURE

Case:	P1	P2	P3	P4	P5
Benchmark	3.27	2.23	2.86	1.97	2.4
Heuristic	2.42	1.98	2.22	1.43	2.05
Improved %	25.99	11.21	22.38	27.41	14.58

TABLE IV
COMPARISON ON MRWJ MEASURE

Case:	P1	P2	P3	P4	P5
Benchmark	7.13	4.74	5.38	4.7	5.8
Heuristic	6.58	4.54	5.26	3.67	5.69
Improved %	7.71	4.22	2.23	21.91	1.90

The computing time of the algorithm is less than a second including the initialization stage. Rapid generation of the solution by the computerized system will encourage expert train planners to develop more scenarios in connection with a preliminary plan and change assumptions to enhance the output of the system.

ACKNOWLEDGMENT

The authors would like to thank RAJA Passenger Train Operating Company, which is affiliated with Iran's Railway Company, RAI, and particularly S. Afzalnia for supplying data and information required for this work.

REFERENCES

- [1] P. Baptiste and C. Le Pape, "Disjunctive constraints for manufacturing scheduling: Principles and extensions," in *Proc. 3rd Int. Conf. Computer Integrated Manufacturing*, Singapore, 1995.
- [2] M. Carey, "A model and strategy for train pathing with choice of lines, platforms, and routes," *Transp. Res.-B*, vol. 28B, no. 5, pp. 333–353, 1994.
- [3] C. K. Chiu, C. M. Chou, J. H. M. Lee, H. F. Leung, and Y. W. Leung, "A constraint-based interactive train rescheduling tool," in *Proc. 2nd Int. Conf. Principle and Practice of Constraint Programming-CP'96*, Cambridge, MA, Aug. 1996.
- [4] X. Cai and C. J. Goh, "A fast heuristic for the train scheduling problem," *Comput. Oper. Res.*, vol. 21, no. 5, pp. 499–510, 1994.
- [5] *Generic Regulations of Movement* (in Persian). Tehran, Iran: Islamic Republic of Iran's Railway Corporation, 1994, vol. 1.
- [6] *Regulations of Movement for Interlocking, Remote Control, and Centralized Traffic Control Systems* (in Persian). Tehran, Iran: Islamic Republic of Iran's Railway Corporation, 1992, vol. 1.
- [7] C. Le Pape, "Implementation of resource constraints in ILOG SCHEDULE: A library for development of constraint-based scheduling systems," *Intell. Syst. Eng.*, vol. 3, no. 2, pp. 55–66, 1994.
- [8] —, "Three mechanisms for managing resource constraints in a library for constraint-based scheduling," in *Proc. INRIA/IEEE Conf. Emerging Technologies and Factory Automation*, Paris, France, 1995.
- [9] A. I. Mees, "Railway scheduling by network optimization," *Math. Comput. Modeling*, vol. 15, no. 1, pp. 33–42, 1991.
- [10] E. R. Peterson and J. Taylor, "A structured model for rail line simulation and optimization," *Transp. Sci.*, vol. 16, no. 2, pp. 192–206, 1982.
- [11] B. Stroustrup, *The C++ Programming Language*, 2nd ed. Reading, MA: Addison-Wesley, 1991.
- [12] P. Van Hentenryck and C. M. Teng, "A generalized arc-consistency algorithm and its specialization," *Artif. Intell.*, vol. 57, no. 3, pp. 291–321, 1992.



Mohammad T. Isaai was born in Tafresh, Iran, in September 1965. He received the B.Sc. and M.Sc. degrees in industrial engineering in 1989 and 1991, respectively, from Sharif University of Technology, Tehran, Iran. He is currently pursuing the Ph.D. degree at the University of Manchester Institute of Science and Technology, Manchester, U.K.

From 1989 to 1991, he was with Nazm-Avaran Management Consulting Company as a System Analyst working on the strategic planning project using simulations for Iran's railway company, RAI.

He started his work with RAI as a Consultant Engineer in 1991, then as a member of the productivity enhancement committee. He became Coordinator of a project, done by DPI, to design and implement decision-support tools and a data-gathering network for the scheduling of trains, crews, and rolling stock. His research interests are decision-support systems, intelligent search techniques, and optimization.



Madan G. Singh (SM'82–F'89) received the B.Sc. degree from Exeter University, U.K., the M.Sc. degree from Manchester University, U.K., the Ph.D. degree from Cambridge, U.K., and the Docteur es Sciences from the University of Toulouse, France.

He is the Co-Founder and Deputy Chairman of Knowledge Support Systems (KSS) Group, plc. KSS provides services related to the intelligent tactical decision-support systems technology developed by Professor Singh and his collaborators on Pricing in Mass Retailing, Petroleum, Banking, Telecommu-

nications, and other industries and for Advertising Budget Allocation, Shelf Space Allocation in Retailing, etc. His consultancy clients have included: Midland Bank, Lloyds Bank, Texaco, Shell, and Unilever. KSS is currently enlarging its capital via a private placing prior to a flotation in 1999 on the London Stock Exchange. He has edited the 18 volume *Encyclopedia of Systems and Control*, and has authored or coauthored 18 books and more than 170 scientific articles.

Dr. Singh holds the Chair of Information Engineering at the University of Manchester Institute of Science and Technology (UMIST), U.K., where he has held a Chair since 1979. As an international leader in the field of Systems Engineering and Decision Support Technology, he was elected the first non-American President of the IEEE Systems, Man, and Cybernetics Society (1994, 1995) and won the 1993 Norbert Weiner Award for truly outstanding contributions to research and scholarship in Information and Decision Technologies, Optimization and Control and Systems Engineering. In February 1994, he was made a Chevalier dans l'ordre des Palmes Academiques by a decree signed by the Prime Minister of France and was graciously granted unrestricted permission by Her Majesty The Queen to wear the insignia of this honor. He was awarded the degree of Doctor of Engineering, honoris causa, by the University of Waterloo, Canada in 1996.